An Attention-Based Lattice Network for Hyperspectral Image Classification

Mohammad Nikzad, *Student Member, IEEE*, Yongsheng Gao, *Senior Member, IEEE*, and Jun Zhou, *Senior Member, IEEE*

***Abstract*— Convolutional neural networks (CNNs) with 3-D convolutional kernels are widely used for hyperspectral image (HSI) classification, which bring notable benefits in capturing joint spectral and spatial features. However, they suffer from poor computational efficiency, causing the low training/inference speed of the model. On the contrary, CNN-based methods with 1-D and 2-D kernels are efficient but mostly restricted to extracting either spectral or spatial features. Moreover, most CNN-based HSI classification frameworks are incapable of simultaneously taking advantage of residual and dense aggregations without over-allocating parameters or information loss for feature reusage. This article presents a novel attention-based lattice network (ALN) to overcome these shortcomings. The proposed 2-D lattice framework can effectively harness the advantages of residual and dense aggregations to achieve outstanding accuracy performance and computational efficiency simultaneously. Furthermore, the ALN employs a unique joint spectral–spatial attention mechanism to capture both spectral and spatial information effectively. In particular, a new pointwise spectral attention mechanism is adopted to fully capture spectral dependences for every pixel. Our extensive experimental investigation verifies the effectiveness and efficiency of the ALN architecture for HSI classification.**

***Index Terms*— Attention-based lattice network (ALN), convolutional neural networks (CNNs), deep learning, hyperspectral image (HSI) classification.**

I. Introduction

The hyperspectral image (HSI) analysis plays a fundamental role in a wide range of remote sensing applications, such as environmental management [1] and agriculture [2], [3]. HSIs are captured from hyperspectral sensors and consist of many narrow spectral bands that carry detailed spectral and spatial information of the observed objects. With the capability to support discriminative spectral and spatial features, the analysis makes HSI a favorable choice of precise identification and characterization of the physical, chemical, and material properties of the areas of interest.

Manuscript received October 30, 2021; revised February 12, 2022 and March 26, 2022; accepted March 28, 2022. Date of publication April 4, 2022; date of current version April 18, 2022. *(Corresponding author: Yongsheng Gao.)*

Mohammad Nikzad and Yongsheng Gao are with the School of Engineering and Built Environment, Griffith University, Nathan, QLD 4111, Australia, and also with the Institute for Integrated and Intelligent Systems, Griffith University, Nathan, QLD 4111, Australia (e-mail: n.nikzaddehaji@griffith.edu.au; yongsheng.gao@griffith.edu.au).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia, and also with the Institute for Integrated and Intelligent Systems, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

Digital Object Identifier 10.1109/TGRS.2022.3164341

However, the extensive number of spectral bands may cause feature redundancy and the curse of dimensionality problems [4]. Much effort has been made in the literature to alleviate the challenges in using band selection or feature extraction approaches [5]–[7]. In addition to spectral information, as there is spatial consistency in the HSI, i.e., pixels within an adjacent spatial area tend to have the same class label, exploiting spatial correlation has also been studied. In this regard, methods such as sparse representation [8], [9] and superpixel-based approaches [10], [11] have taken advantage of spatial information to enhance HSI classification performance. Moreover, as taking advantage of both spectral and spatial information together is desirable, methods including 3-D local binary pattern (LBP) [12] and 3-D Gabor wavelet transformation [13] are applied to exploit spatial–spectral characteristics of HSIs.

The traditional approaches rely on handcrafted features and cannot effectively extract discriminative features for robust HSI classification. On the other hand, convolutional neural networks (CNNs) have shown outstanding performance on many computer vision and image processing tasks [14]. Inspired by the biological process of the visual system, they produce discriminative and representative features from raw input signals without any prior manipulation. Thus, CNNs effectively address the limitation of conventional HSI classification methods. They deploy a unified end-to-end learning-based framework to optimize feature extraction tasks as well as classification tasks simultaneously. In particular, residual networks [15] and densely connected convolutional networks [16] are two major advances to facilitate information circulation and the ability to explore effective new features. Various 1-D and 2-D CNN-based methods are adopted to extract the spectral and/or spatial features, respectively. For instance, Hu *et al.* [17] developed a deep convolutional network that employs 1-D convolution functions to exploit the spectral feature of each pixel's spectrum. In [18], a unified framework is proposed, which integrates 2-D CNN and recurrent neural network (RNN) to acquire both spectral and spatial information. However, many recent CNN-based methods employ 3-D convolutional kernels to exploit both spatial and spectral information at the same time [19]–[22], so that rich spectral–spatial- features can be generated for HSI classification.

Despite the significant improvement of different CNN-based frameworks over traditional approaches for HSI classification, they suffer from several shortcomings. First, 1-D CNNs and

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

2-D CNNs are not capable of considering both spectral and spatial properties and are limited to either spectral-based or spatial-based classification [22], [23]. The 3-D CNNs can address this drawback, however, with an expensive computational overhead [24]. Second, due to the small training sample issue of HSI datasets, many existing CNN-based HSI classification methods cannot effectively and yet efficiently take advantages of residual [15] and/or dense [16] information aggregation advances so that they may be threatened by overfitting issue and lack of good generalization capability.

Recently, we [25] proposed a method to estimate the clean speech magnitude, which is corrupted by different noise levels. This method used a residual-dense lattice (RDL) topology to exploit the advantages of both residual and dense aggregations without over-allocating parameters for feature reusage [26]. This deep lattice framework showed the state-of-the-art performance in enhancing sequential signals, e.g., speech. Despite the success of the lattice topology for speech enhancement, the existing lattice structure introduced in [25] is not applicable for processing the 3-D spectral–spatial hyperspectral cubes. Moreover, RDL networks (RDL-Nets) [25] are casual architectures for sequential signal processing in which the output at time t is only over time steps up to t, not $t + 1$.

In this article, we introduce a new network structure to fully exploit the potential of lattice topology for addressing the aforementioned shortcomings of CNN-based HSI classification methods. To this end, first, we propose a unique triangular lattice (TL) topology of 2-D convolutional kernels, which provides a lightweight model empowered by both residual and dense aggregation links. The TL structure can effectively tackle extensive computation drawbacks of 3-D CNNs for HSI classification. Moreover, different from the topology in RDL-Nets [25], here, a two-step bottleneck technique is incorporated with each convolutional element. Also, the new TL topology is formed by only one TL of convolutional units instated of two lattices of RDL structure [25], [27]. These two features bring more computational efficiency to the model. We further design a lightweight and yet effective joint spectral–spatial (JSS) attention mechanism to the proposed HSI classification framework to effectively exploit information carried by interdependences among spectral bands and spatial features. The proposed JSS attention mechanism can successfully address the limitation of 1-D CNNs and 2-D CNNs in considering HSIs' spectral and spatial properties, simultaneously. We name our proposed network, built from the new TL architecture, as an *attention-based lattice network* (ALN), whose overview is illustrated in Fig. 2. Unlike the RDL-Nets presented in [25], the proposed ALN is a noncasual method designed for spatial signal processing (e.g., image processing).

The main contributions of our work are summarized as follows.

1) First, the proposed 2-D convolutional lattice can provide a lightweight framework to address the small training sample set issue in remote sensing HSI classification and maintain the efficiency of the ALN model in terms of computation cost and time.
2) Second, the proposed two-stage attention strategy ensures that the JSS features are exploited throughout the network. In particular, the first stage of attention mechanism aims at recalibrating the raw input data cube based on its spectral spectrum and spatial positions. The second gating stage is responsible for refining the final extracted feature maps.
3) Third, extensive experiments show that the new ALN can produce higher HSI classification performance than several benchmark CNN-based HSI classification methods without sacrificing efficiency in speed, model size, and computational cost. Furthermore, extensive ablation analyses are conducted to investigate the role of every component of the proposed network.

The rest of this article is organized as follows. Section II briefly reviews related works. The proposed JSS attention mechanism and ALN framework are presented in Sections III and IV, respectively. The HSI datasets used in this study, ablation analysis, and the experimental results are provided in Section V. Conclusions are summarized in Section VI.

II. Related Works

A. Deep Learning for Hyperspectral Image Classification

In the past few years, significant progress has been made for HSI classification by developing discriminative learning-based methods using deep CNNs. Generally, CNN-based HSI classification methods are categorized into three main groups: spectral CNNs, spatial CNNs, and spectral–spatial CNNs. While the first group focuses on extracting spectral information for each pixel in HSIs, the second group relies on extracting spatial correlation of pixels within a spatial region. The third group of CNN-based HSI classification methods aims at integrating both spectral and spatial features simultaneously. Since each spatial pixel in the HSI corresponds to a spectral vector with hundreds of bands, 1-D CNN-based methods are widely used to exploit a deep feature of the spectrum [17]. In addition to popular 1-D CNN frameworks, 1-D generative adversarial networks (GANs) [28] are also applied to extract spectral information.

With respect to the spatial-based classification approaches, 2-D CNNs are the most common methods used for RGB image analysis and recognition tasks. Chen *et al.* [29] proposed the very first 2-D CNN model, which extracts deep features from the latent space of a stacked autoencoder framework to classify hyperspectral data. Likewise, a stacked sparse autoencoder (SSAE) [30] was introduced to exploit a multiscale sparse deep spatial feature representation for HSI classification. In general, before feeding the HSI into the 2-D CNN, it is preprocessed via principal ccomponent analysis (PCA) to select the most informative spectral bands [31]. Deep belief networks (DBNs) alongside active learning are also employed by Liu *et al.* [32] to obtain an effective classification framework.

The 3-D CNN architectures are mostly employed to extract JSS information [33], [34]. The 3-D deep framework proposed by Hamid *et al.* [22] takes a 3-D input cube to extract spatial and spectral features simultaneously. Li *et al.* [23] also adopted 3-D convolutional kernels to integrate spectral and spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

information together. Using 3-D CNNs to extract multiscale spatio-spectral relationships in [21] could add further improvement to 3-D CNN-based approaches. However, these 3-D convolutional methods impose high computational costs and parameter size overhead [33], [35]. To overcome this issue, many recent works aim to utilize two-branch topologies in which one of the branches is responsible for spectral features extraction, and the other focuses on spatial information. For instance, in [36] and [37], parallel 1-D CNN and 2-D CNN streams were proposed to extract spectral and spatial features, respectively. The results of two branches are then fused using different strategies such as the weighted summation method [24]. However, combining the results of networks' spectral and spatial branches still remains as a challenging unsolved problem.

In this article, different from the aforementioned CNN-based approaches, we design a novel 2-D CNN topology that can effectively and efficiently exploit spectral and spatial hyperspectral features simultaneously through only one streamline of information.

B. CNN Architectures

Over the last decade, various CNNs such as AlexNet [38], Visual Geometry Group (VGG) [39], and GoogleNet [40] have shown outstanding performance in computer vision and image classification [14], [41]. Residual (ResNets) [15] and dense (DenseNets) [16] aggregations of layer outputs have presented further noticeable progress in the development of CNNs' topologies. Residual links facilitate passing information and gradients efficiently through the network. Dense aggregations offer direct feature reusage, as deeper layers have access to the outputs of shallower layers. In fact, they are able to take advantage of very deep network structures by applying shortcut connections between stacked layers. In this regard, Paoletti *et al.* [42] proposed a deep pyramid residual network (DPyResNet) for HSI classification. Likewise, the spectral–spatial residual network (SSRN) [19] took advantages of the residual aggregations' technique to improve HSI classification performance. Wang *et al.* [43] proposed an end-to-end fast dense spectral–spatial convolution (FDSSC) framework for HSI classification. The FDSSC framework uses different convolutional kernel sizes to extract spectral and spatial features separately. Kang *et al.* [44] combined both residual and dense shortcut connections to design a dual-path network (DPN) for effectively extracting the discriminative features from the complex scene in HSIs.

Despite the success of residual and dense skip connections in HSI classification, both aggregation types have drawbacks. As shown in [25] and [26], these techniques may cause information loss and/or parameter over-allocating as the networks get deeper or wider. Huang *et al.* [45] used a stochastic residual strategy to alleviate information loss issues related to the residual networks. More specifically, since there often exist small numbers of training samples for HSI classification, using generic residual and/or dense topologies may result in overfitting issue and poor generalization capability. In [35], a lightweight CNN for HSI classification was presented by taking aid from spatial–spectral Schroedinger eigenmaps (SSSE) [46] operator, and a dual-scale convolution procedure (DSC).

C. Attention Networks

The critical role of a visual attention mechanism in human perception has been shown through a range of studies [47], [48]. The human vision system efficiently and adaptively focuses on salient areas and suppresses other irrelevant information. An attention mechanism has been widely incorporated in computer vision to improve the efficiency of the model [49], [50]. The residual attention network [51] introduces a noise-resistant encoder–decoder style module to generate a 3-D attention map for refining the intermediate feature maps. However, this attention framework suffers from high computational/parameter overhead.

Recently, Hu *et al.* [52] proposed a lightweight gating mechanism to exploit interchannel relationships of the convolutional feature maps. Their squeeze-and-excitation (SE) module measures channelwise attention by squeezing global spatial information. Following the idea of squeeze and excitation networks (SENets') attention mechanism, spectral attention (also known as channelwise attention) and spatial attention are two main streams that have been targeted for attention module advances in the HSI domain [53], [54]. Since the final purpose of HSI classification is to categorize the input pixel into a relevant class, pixelwise exploitation of interband relationships will result in a fine-grained spectral gating mechanism. Common attention mechanisms such as SENets [52], bottleneck attention module (BAM) [54], and convolutional block attention module (CBAM) [53] use global max and/or average pooling to summarize information carried by every feature map. The global pooled information is not capable of inferring pixelwise channel attention. However, as shown in Fig. 1, the proposed JSS technique obtains a pointwise spectral attention map by directly applying fully connected (FC) layers on every pixel.

Wang *et al.* [55] proposed a band attention network for HSI classification. Zhu *et al.* [56] introduced a spectral–spatial attention layer to SSRN [19]. Similar to SENets, Roy *et al.* [57] introduced FuSENet for HSI classification by combining different squeezing techniques [i.e., global average pooling (GAP) and global max pooling]. On the other hand, Fang *et al.* [58] incorporated a spectralwise attention mechanism into a dense convolutional network (MSDNSA). More recently, a two-branch spectral–spatial network named spatial–spectral attention-aided CNN (SSAtt) was proposed for HSI classification [24]. However, the two subnetworks have to be trained separately and then the whole network should be trained again for being fine-tuned. Moreover, an attention-based adaptive spatial–spectral kernel-improved residual network (A2S2K-ResNet [33]) adopted selective 3-D convolutional kernels to construct a spectral–spatial attention mechanism. Due to 3-D convolutional kernels, A2S2K-ResNet [33] model is relatively complex and slow compared with 2-D models.

III. Joint Spectral–Spatial Attention Mechanism

Capturing joint contextual and spectral information is desired for enhancing the performance of the CNN-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

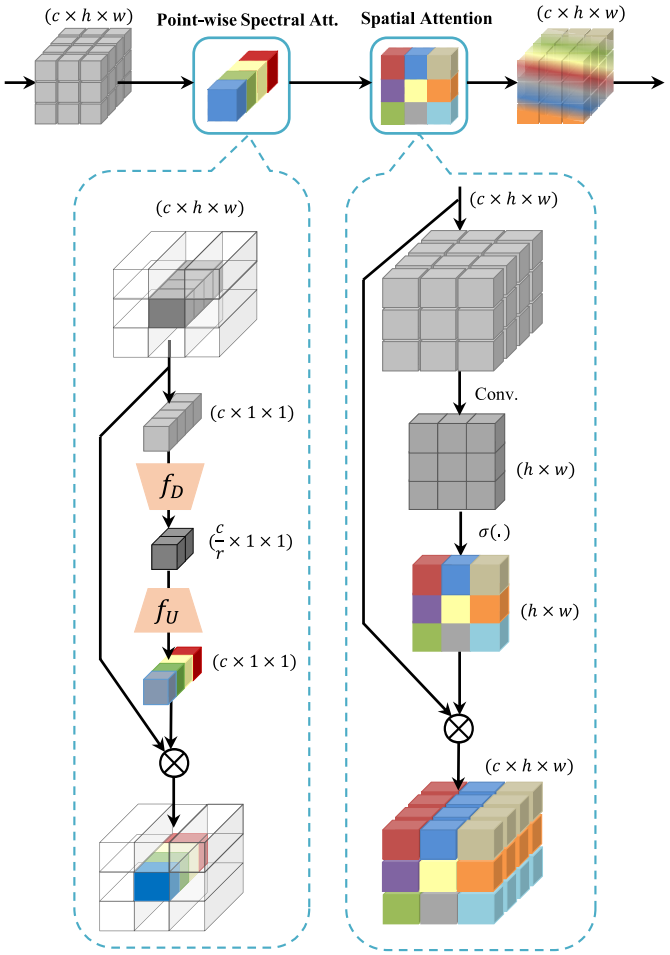


Fig. 1. Overview of the proposed JSS attention module. The left and right dashed boxes illustrate the details of the pointwise spectral and spatial attention modules, respectively. ⊗ represents elementwise multiplication operator. f_D and f_U indicate shared FC layers for pointwise channel reduction and up-sampling, respectively.

HSI classification methods [22], [23]. Using 3-D convolutional layers is a common way to exploit both spatial and spectral information. However, 3-D CNNs can cause a significant increase in the number of parameters and computation cost [24]. Recently, Hang *et al.* [24] proposed an attention-aided 2-D CNN framework in which a spectral attention subnetwork and a spatial subnetwork are responsible for extracting spectral and spatial features, respectively. Unlike this strategy, we leverage joint spectral and spatial attention mechanisms into a sequential design of 2-D convolutional elements. The idea behind the new JSS attention module is illustrated in Fig. 1. The input 3-D tensor is refined by applying two consecutive pointwise spectral and spatial attention modules, respectively.

Typically, all channelwise features are treated equally by CNN-based HSI classification networks [22], [23], [42]. This policy hinders the network's ability to deal with different types of spectral information. Hence, considering the interdependences among bands (channels) as a spectral attention mechanism can achieve a better classification performance [52], [55]. Moreover, it is well-known that utilizing contextual information is crucial to know which spatial locations should be focused on [53]. In this article, both spectral and spatial axes of attention are exploited with a simple yet effective JSS attention module. As shown in Fig. 1, the JSS attention module ρ_{JSS} refines a given intermediate feature map $x \in \mathbb{R}^{c\times h\times w}$ sequentially on its spectral bands and spatial locations, respectively. The recalibrated feature map $\bar{x}$ is computed as

$$\bar{x} = \rho_{\text{JSS}}(x) = \rho_s\left(\rho_{\text{pw}}(x)\right) \tag{1}$$

where $\rho_s(.)$ and $\rho_{\text{pw}}(.)$ represent spatial and pointwise spectral attention modules, respectively.

1) Pointwise Spectral Attention: Using squeeze and excitation operations [59] is a common way to adopt a spectral (channelwise) attention mechanism. To this end, they generally apply a global pooling operator (e.g., average) to shrink the given c feature maps through their spatial dimensions $h \times w$. Despite the advantage of these methods, the global pooled information is not the optimal features to infer accurate channel attention [53]. Furthermore, since the final goal is pixelwise classification of HSI, global information may impact the performance. To address this shortcoming, we remove the squeezing function (global pooling) and adopt a pointwise excitation mechanism to fully capture spectral dependences for every pixel. The structure of the pointwise spectral attention module is illustrated in Fig. 1. The excitation operator acts as a gating mechanism to learn nonlinear synergies between channels alongside a nonmutually exclusive relationship for every spatial location. Here, we adopt the pointwise spectral gating mechanism by forming a shared bottleneck with two FC layers. The first FC layer followed by rectified linear unit (ReLU) [60] nonlinearity (δ) reduces the spectrum dimension with ratio r (here, we set $r = 16$), and then, an up-sampling FC layer followed by sigmoid activation (σ) returns the input spectral dimension (c). Therefore, the final pointwise spectral attention map M_{pc} can be formalized as

$$M_{\text{pw}}(i, j) = \sigma\left(f_U(\delta(f_D(x(i, j))))\right) \tag{2}$$

where $x(i, j) \in \mathbb{R}^{c\times 1\times 1}$ refers to the spectral feature vector at position (i, j). f_D and f_U represent the shared FC layers for channel reduction and up-sampling operators, respectively. Finally, refined input cube is adaptively formed by M_{pw} as

$$\rho_{\text{pw}}(x) = M_{\text{pw}} \otimes x \tag{3}$$

where $\otimes$ represents the elementwise multiplication operator.

2) Spatial Attention: As shown in Fig. 1, the spatial attention module is constructed by projecting the feature $x \in \mathbb{R}^{c\times h\times w}$ into a reduced spectral dimension $\mathbb{R}^{1\times h\times w}$ using a 1×1 convolution. Then for spatial refinement, a sigmoid activation function (σ) is applied to refine contextual information effectively. In short, the spatial attention map M_s is computed as

$$M_s = \sigma\left(f^{1\times 1}(x)\right) \tag{4}$$

where f and superscripts denote convolution operation and convolutional kernel sizes, respectively. The spatially recalibrated feature can be obtained as

$$\rho_s(x) = M_s \otimes x \tag{5}$$

where $\otimes$ represents the elementwise multiplication operator.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

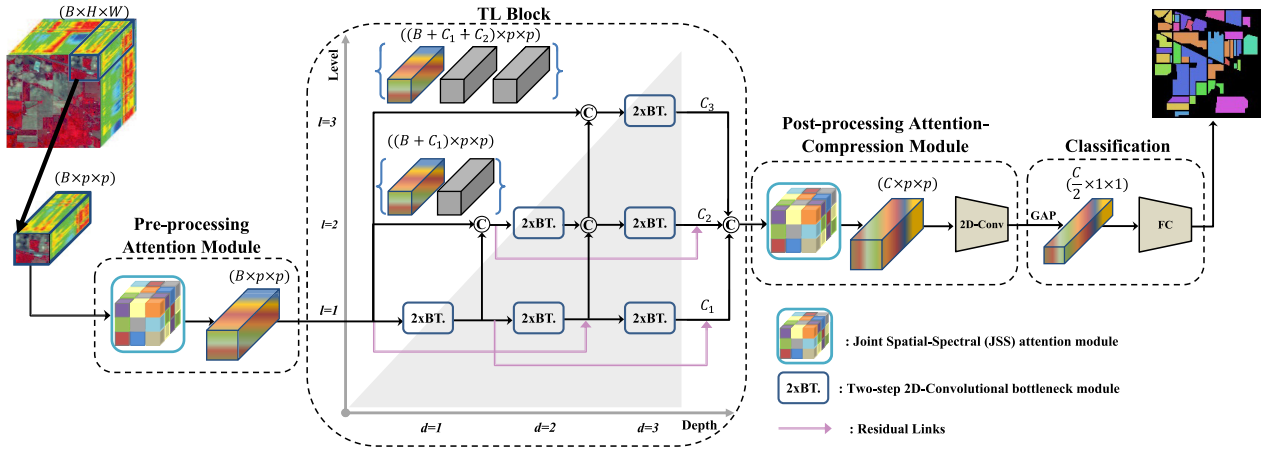


Fig. 2. Overview of the proposed ALN for HSI classification. © indicates the concatenation operator.

IV. Proposed Attention-Based Lattice Network

The proposed ALN is shown in Fig. 2. The proposed HSI classification framework consists of four main components: the preprocessing attention module, the TL block, the postprocessing attention-compression module, and the classification module, respectively. Let $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ be the raw input HSI with height H, width W, and the number of spectral bands B. All spectral vectors (pixels) $\mathbf{x}_{i,j} \in \mathbb{R}^B$ (with $i = 1, \ldots, H$, and $j = 1, \ldots, W$) are categorized into N land-cover categories $\mathbf{z}_n$ (with $n = 1, \ldots, N$). The joint spatial and spectral 3-D cube $\mathbf{x} \in \mathbb{R}^{B\times p\times p}$ is used by extracting the $p \times p$ neighborhood region centered at pixel (i, j) from the original image $\mathbf{X}$. Then, the cube is fed into ALN to obtain classification result.

A. Preprocessing Attention Module

Due to the high spectral dimensionality of real-world hyperspectral data samples, their information redundancy and noisy spectral bands can affect the performance of the HSI analysis system. To address this issue, various dimensionality reduction methods, such as PCA [6]–[61] and spatial–spectral Schrodinger eigenmaps (SSSE) [35], [46], have been used as the preprocessing step. However, these methods may suffer from information loss. In this work, we incorporate a spatial–spectral gating mechanism to emphasize salient parts of data samples rather than applying pruning approaches [6]–[61]. As shown in Fig. 1, the input cube $\mathbf{x}$ is sequentially refined through two consecutive pointwise spectral and spatial attention modules, respectively

$$\bar{\mathbf{x}} = \rho_{\text{JSS}}(\mathbf{x}) = \rho_s\big(\rho_{\text{pw}}(\mathbf{x})\big). \tag{6}$$

B. Triangular Lattice Block

Our proposed lattice is a TL of convolutional units. The location of each convolutional unit, C_{ld}, within the lattice is specified by level and depth coordinates (l, d)

$$C_{ld} = \begin{cases} C_{ld}^{\triangle}, & \text{if } l \le d,\ d \le D \\ \emptyset, & \text{otherwise} \end{cases} \tag{7}$$

where $l = 1, 2, \ldots, L$ and $d = 1, 2, \ldots, D$. The maximum level (height) of the lattice is L, and the depth is D. $C_{ld}^{\triangle}$ is a convolutional unit in the lattice triangle. $\emptyset$ indicates that the convolutional unit does not exist. Local dense aggregations of convolutional unit outputs are formed strictly over the *height* of the lattice. While the lattice block does not allow for total feature reusage, densely aggregating only a subset of previous outputs is beneficial [25]. Local residual (LR) links are also adopted to improve intrablock gradient flow. The proposed TL topology consists of two main parts: local dense and LR aggregations. They are explained in detail in Sections IV-B1 and IV-B2. To bring more computational efficiency, we use two-step bottleneck layers. This technique is widely used by CNN architectures such as ResNets [62] and DenseNets [16] and helps restrain the number of parameters, as the number of convolutional units is increased. Thus, here, each convolutional unit has a two-layer bottleneck design. Each layer is a preactivated composite function consisting of three operations, including ReLU activation (δ) [63], batch normalization (**BN**) [64], and followed by 2-D convolution (f). The first layer has a kernel and output size of $k = 1 \times 1$ and (c/r) (in this article, we set $r = 16$), respectively. The second layer uses a kernel size of $k = 3 \times 3$ and an output size of c. In short, the spatially refined output of the two-step convolutional module y is computed as

$$\begin{aligned} \hat{x} &= f^{1\times 1}(\mathbf{BN}(\delta(x))) \\ y &= f^{3\times 3}(\mathbf{BN}(\delta(\hat{x}))) \end{aligned} \tag{8}$$

where f and superscripts denote convolution operation and convolutional kernel sizes, respectively. y denotes the final output of the convolutional unit.

1) Local Dense Aggregations: The input to a convolutional unit in the lattice, x_{ld}, is the dense aggregation of the outputs at depth $d - 1$, and levels $l, l - 1, \ldots, 1$

$$x_{ld} = \begin{cases} \bar{\mathbf{x}}, & \text{if } l = d = 1 \\ y_{l(d-1)}, & \text{if } d > 1, l = 1 \\ \left[y_{l(d-1)}, x_{(l-1)d}\right], & \text{if } d > l, l > 1 \\ \left[\bar{\mathbf{x}}, x_{(l-1)d}\right], & \text{if } d = l, l > 1 \end{cases} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

where [.] denotes the concatenation operation, and y_{ld} is the LR aggregation at (l, d). The local dense aggregations in the lattice allow for multiple concise outputs to be progressively formed. By densely aggregating outputs over the height of the lattice, the input size to deeper convolutional units within the block is also limited and avoids over-parameter allocating.

2) Local Residual Aggregations: To improve the flow of gradients over the length of the lattice, LR links are adopted

$$y_{ld} = \begin{cases} y_{ld} + x_{l(d-1)}, & \text{if } d > l \\ y_{ld}, & \text{if } d \leq l. \end{cases} \tag{10}$$

When the sizes of y_{ld} and $x_{l(d-1)}$ are nonidentical, the residual link is weighted so that $x_{l(d-1)}$ is of the same size as y_{ld}. The final output of the lattice y_{TL} is formed by concatenation of all levels of residual outputs generated at depth D (as shown in Fig. 2)

$$y_{\text{TL}} = [y_{1D}, y_{2D}, \ldots, y_{LD}] \in \mathbb{R}^{C \times p \times p} \tag{11}$$

where [.] represents the concatenation operation. $C = \sum_{l=1}^{L} C_i$ in which C_i denotes the convolutional elements' output size at each level of the lattice.

C. Postprocessing Attention-Compression Module

Since the output of the lattice block contains multilevel features, similar to the preprocessing stage (Section IV-A), we apply JSS attention on y_{TL}

$$\bar{y}_{\text{TL}} = \rho_{\text{JSS}}(y_{\text{TL}}) = \rho_s\big(\rho_{\text{pw}}(y_{\text{TL}})\big). \tag{12}$$

In order to integrate and compress the refined feature map $\bar{y}_{\text{TL}} \in \mathbb{R}^{C \times p \times p}$ across the channel dimension, it is projected into the reduced spectral dimension $\mathbb{R}^{(C/2) \times p \times p}$ using 1×1 convolution ($f^{1\times1}$) followed by BN and ReLU activation function (δ):

$$\mathbf{y} = \delta\big(\mathbf{BN}\big(f^{1\times1}(\bar{y}_{\text{TL}})\big)\big). \tag{13}$$

D. Classification

After the attention-compression block, a GAP operator is used to transform 3-D feature maps $\mathbf{y}$ of size $C \times p \times p$ into a feature vector of size $(C/2) \times 1 \times 1$. Then, a softmax-based FC layer is used to produce the category probability $\hat{\mathbf{z}} \in \mathbb{R}^{N \times 1}$

$$\hat{\mathbf{z}} = \text{softmax}(f_N(\text{GAP}(\mathbf{y}))) \tag{14}$$

where f_N represents the final FC layer. Finally, we adopt the following loss function to optimize the model:

$$\mathcal{L}(\Theta) = -\sum_{n=1}^{N} \mathbf{z}_n \log(\hat{\mathbf{z}}_n) \tag{15}$$

where Θ indicates the trainable parameters in ALN.

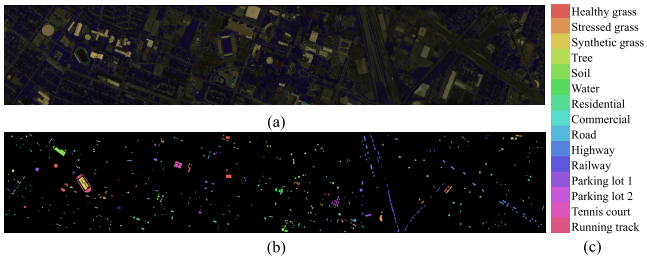


Fig. 3. (a) False-color image. (b) Ground-truth map. (c) Land-cover types of the Houston-2013 dataset.

E. Implementation Details

In this work, we set the convolutional unit's output size at each level to $C_l = C_{l-1} + 32$, where $C_1 = 64$ is the output size at $l = 1$. This setting ensures that a reduced number of parameters are used for feature reusage. The maximum depth and levels of the lattice block are set to $D = 3$ and $L = 3$, respectively. Concerning the spatial size of the input 3-D cube ($p \times p$), patches with the resolution of 9×9 are used.

We use stochastic gradient descent (SGD) for the training of the proposed models. In particular, we adopt the Adam optimizer [65] with default hyperparameters. All models are trained for 200 epochs with a mini-batch size of 128 and an initial learning rate of 0.001, which is decayed to 10^{-6} by adopting the cosine annealing technique [66]. A single NVIDIA Geforce Titan RTX graphics processing unit (GPU) card is used to carry out all experiments.

V. Experimental Results and Discussion

The effectiveness of the proposed ALN for HSI classification is evaluated across three well-known benchmark hyperspectral datasets. Furthermore, we use overall accuracy (OA), average accuracy (AA), and kappa (κ) statistics to evaluate the performance. Specifically, OA is defined as the ratio of correctly classified samples among the total test samples. AA represents the average of classwise accuracy, and κ measures strong mutual agreement between the given ground-truth map and the classification map generated by the model. To avoid the influence of weight initialization and randomness of data sampling, we repeat all experiments ten times and mean values and corresponding standard deviation values are reported.

A. Datasets

1) Houston-2013: This dataset was captured over the University of Houston campus and the neighboring urban area [67] with a ground sampling distance of 2.5 m per pixel. The image data have the spatial dimension of 349×1905 in which pixels are labeled from 15 different classes with 144 spectral bands. On this dataset, the training and testing sets are separately labeled. The detailed pixel distribution per class is reported in Table I. The false color and the training and test maps of the Houston-2013 data are shown in Fig. 3.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PIXEL CLASS DISTRIBUTIONS WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR THE HOUSTON-2013 DATASET

Class No.	Class Name	Training	Test
1	Healthy grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total	-	2830	12197

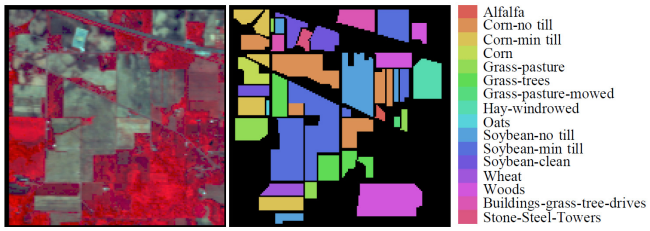


Fig. 4. (a) False-color image. (b) Ground-truth map. (c) Land-cover types of Indian Pines dataset.

TABLE II

PIXEL CLASS DISTRIBUTIONS WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR THE INDIAN PINES DATASET

Class No.	Class Name	Training	Test
1	Alfalfa	15	39
2	Corn-no Till	15	1419
3	Corn-min Till	15	819
4	Corn	15	219
5	Grass/Pasture	15	482
6	Grass/Tree	15	732
7	Grass/Pasture-mowed	15	11
8	Hay-windrowed	15	474
9	Oats	15	5
10	Soybean-no Till	15	953
11	Soybean-min Till	15	2453
12	Soybean-clean Till	15	599
13	Wheat	15	197
14	Woods	15	1279
15	Buildings-Grass-Tree-Drives	15	365
16	Stone-steel-towers	15	80
Total	-	240	10126

2) Indian Pines: This HSI dataset was acquired using an airborne visible/infrared imaging spectrometer (AVIRIS) sensor over agricultural areas in North Western Indiana, USA, with 20 m per pixel spatial resolution. It contains 10 366 labeled pixels distributed in 16 ground-truth classes. Furthermore, the captured scene has a spatial dimension of 145 × 145 with 224 spectral bands. After removing 4 zero bands and 35 lower SNR bands corrupted by the atmospheric water absorption effect, the remaining 185 bands are preserved for classification. Its ground-truth map and detailed information regarding the distribution of samples per class are shown in Fig. 4 and Table II. Following [35], [58], and [68], since the ninth class (Oats) only contains 20 labeled samples, 15 samples are randomly chosen from each category to form the training set.

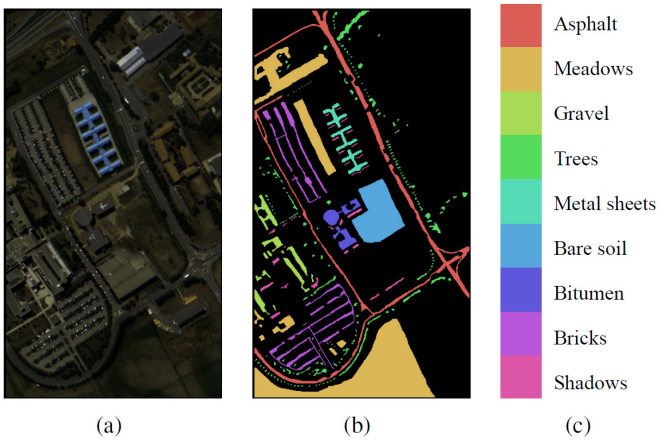


Fig. 5. (a) False-color image. (b) Ground-truth map. (c) Land-cover types of the PaviaU dataset.

TABLE III

PIXEL CLASS DISTRIBUTIONS WITH THE NUMBER OF TRAINING AND TEST SAMPLES FOR THE PAVIA UNIVERSITY DATASET

Class No.	Class Name	Training	Test
1	Asphalt	20	6611
2	Meadow	20	18629
3	Gravel	20	2079
4	Trees	20	3044
5	Metal sheets	20	1325
6	Bare Soil	20	5009
7	Bitumen	20	1310
8	Bricks	20	3662
9	Shadows	20	927
Total	-	180	42596

3) Pavia University (PaviaU): This HSI dataset was recorded by reflective optics system imaging spectrometer (ROSIS) sensor over the university of Pavia, Northern Italy, including nine urban land-cover classes and consisting of 610 × 340 pixels. The raw HSI data contain 115 spectral bands; however, 12 noisy bands are removed, and the remaining 103 spectral channels are used for the experiments. The details of the dataset are listed in Fig. 5 and Table III. Different from the Houston-2013 dataset, Indian Pines and Pavia University datasets do not have predefined training and test sets. Therefore, we randomly select 15 and 20 sample pixels from Indian Pines and Pavia University to form their training sets, respectively, and the remaining pixels are used as their test sets. Furthermore, the input 3-D cube $\mathbf{x} \in \mathbb{R}^{B\times p\times p}$ is formed by extracting the $p \times p$ neighborhood region centered at every training pixel (i, j). Hence, randomly selecting a large number of training cubes will increase the chance of spatial overlapping between input training cube and test pixels. To alleviate this problem and avoid the bias problem as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

ABLATION STUDY OF THE JSS ATTENTION MECHANISM: SPATIAL ATTENTION (SPA-ATT) MECHANISM, POINTWISE SPECTRAL ATTENTION (PSPE-ATT) MECHANISM, PREPROCESSING ATTENTION MODULE (PRE-ATT), AND POSTPROCESSING ATTENTION MODULE (POST-ATT). THE SYMBOLS "✓" AND "✕" INDICATE WHETHER THE CORRESPONDING ELEMENT IS EMPLOYED WITH THE CONFIGURATION OR NOT

Different combinations of Spa-Att, PSpe-Att, Pre-Att and Post-Att.										
PSpe-Att	✕	✓	✓	✓	✕	✕	✕	✓	✓	✓
Spa-Att	✕	✕	✕	✕	✓	✓	✓	✓	✓	✓
Pre-Att	✕	✓	✕	✓	✓	✕	✓	✓	✕	✓
Post-Att	✕	✕	✓	✓	✕	✓	✓	✕	✓	✓
OA(%)	89.24	90.02	89.67	90.16	90.30	90.21	90.48	91.16	90.47	91.68
AA(%)	89.96	90.78	90.57	90.63	90.01	89.92	90.23	91.10	90.82	91.64
κ	0.883	0.891	0.888	0.896	0.895	0.893	0.897	0.908	0.902	0.915

described in [68], we use a small number of training samples on these two Indian Pines and Pavia University datasets.

B. Ablation Study of ALN

This section will conduct an extensive ablation analysis on the effects of two different concepts of the proposed ALN: JSS attention mechanism and TL topology. To this end, different ALN configurations are examined, and the OA, AA, and κ of each configuration on the dataset Houston-2013 are reported.

1) Effects of JSS Attention Mechanism: We comprehensively investigate the influence of the proposed JSS attention mechanism. To this end, we verify the role of the proposed JSS module by examining ten ALN configurations. In particular, ten designs are formed by adding either pointwise spectral attention mechanism (PSpe-Att) or spatial attention mechanism (Spa-Att) to the preprocessing attention module (Pre-Att) and/or postprocessing attention unit (Post-Att). In this experiment, we keep TL block and classification part fixed as described in Sections IV-B and IV-D, respectively.

As shown in Table IV, utilizing either PSpe-Att or Spa-Att to the baseline framework (no PSpe-Att and Spa-Att) results in higher OA and κ values compared with the baseline. The effect of PSpe-Att is separately tested for both preprocessing and postprocessing attention modules. PSpe-Att for preprocessing purpose obtains the OA of 90.02%, while this value for the PSpe-Att at the last gating block is slightly less (89.67%). This suggests that preprocessing of the raw data cube has more influence than the final attention block to improve the performance. Applying the PSpe-Att at both the locations of the ALN leads to further increase in the OA and κ metrics, supporting that both Pre-Att and Post-Att are beneficial.

In addition to PSpe-Att, we carry out a similar investigation as aforementioned to exclusively test the effectiveness of the spatial attention mechanism. Table IV listed the results. Compared with the baseline, Spa-Att can produce above 1% of improvement of OA. Moreover, Spa-Att exhibits a superior performance than the pointwise spectral counterpart module (PSpe-Att). However, they both show a similar trend regarding the position of applying gating blocks.

Finally, the effectiveness of the JSS attention mechanism is evaluated. As shown in Table IV, employing both PSpe-Att and Spa-Att simultaneously can obtain the highest classification performance compared with other analyzed ALN configurations. More specifically, applying the proposed JSS attention mechanism at Pre-Att and Post-Att blocks boosts the OA values of Spa-Att, PSpe-Att, and the baseline configurations by 1.20%, 1.52%, and 2.44%, respectively. Moreover, similar to the aforementioned attention-based configurations, the superiority of the Pre-Att module to the Post-Att block is also verified. The Pre-Att obtains 91.16% of OA, whereas the Post-Att attains 90.47% of overall performance.

TABLE V

ABLATION STUDY ON THE EFFECTS OF DIFFERENT ATTENTION TECHNIQUES

	ALN-SE	ALN-BAM	ALN-CBAM	ALN-JSS
OA(%)	88.77	88.25	88.89	91.68
AA(%)	88.34	88.77	89.92	91.64
κ	0.87	0.87	0.88	0.91

2) JSS Versus Other Attention Techniques: Furthermore, we conduct another ablation study on the role of the different attention mechanisms within the proposed ALN. In particular, we set up four different ALNs configured with the proposed JSS, SE [52], BAM [54], and CBAM [53] attention modules, leading to ALN-JSS, ALN-SE, ALN-BAM, and ALN-CBAM, respectively. We keep all other aspects of the networks fixed as described in Section IV-E for a fair comparison. As shown in Table V, the ALN model empowered with the proposed JSS attention mechanism outperforms other counterparts significantly. This verifies the effectiveness of the pointwise gating mechanism compared with common global-pooled information techniques to exploit pixelwise spectral dependences in HSI classification tasks.

Though the effectiveness of the proposed JSS module has been proved empirically, we also make a deeper investigation to provide a better understanding of the behavior of this part. To this end, we cropped an input 3-D cube $\mathbf{x} \in \mathbb{R}^{144\times 70\times 70}$ [a 70×70 region highlighted by the blue rectangle in Fig. 6(a)] from the Houston-2013. The input cube is fed into the pretrained networks, ALN-JSS, ALN-SE, ALN-BAM, and ALN-CBAM, respectively, and the scaled outputs are generated by the first (preprocessing) attention unit of the networks. Due to the high number of spectral bands (i.e., 144) and for the sake of visualization, the 3-D outputs are reduced

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

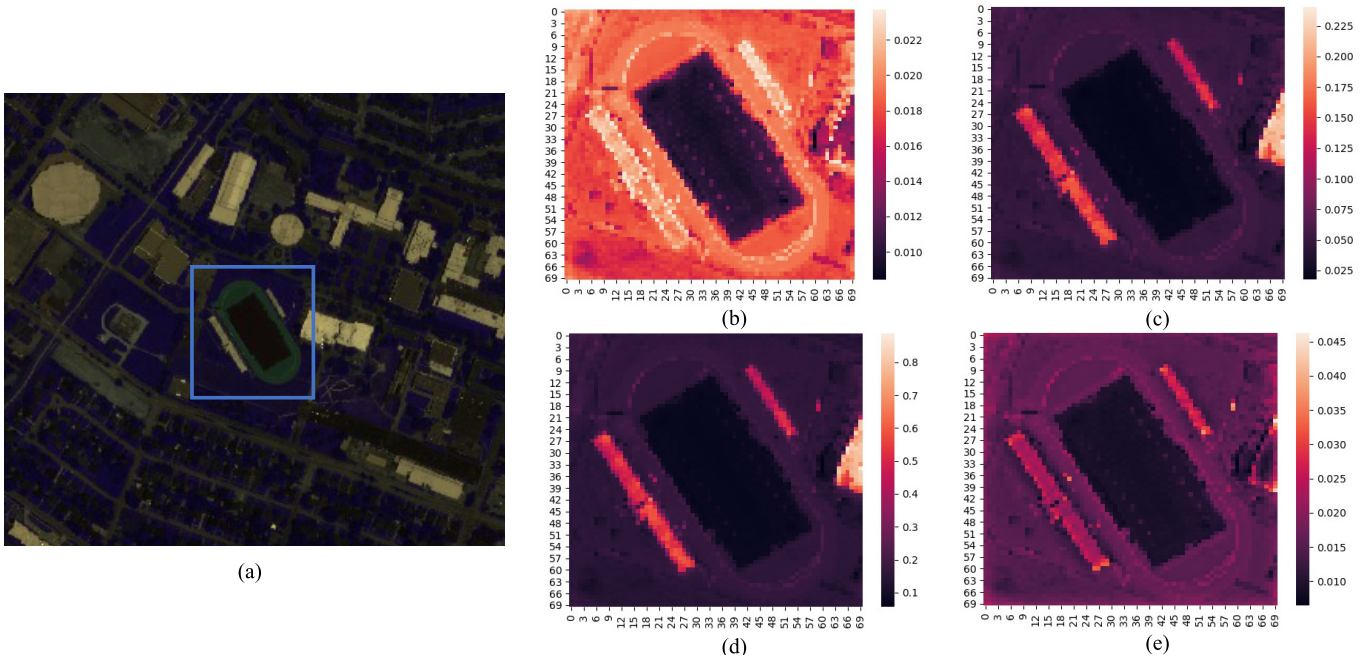


Fig. 6. (a) False-color image cropped from Houston-2013 dataset, (b) corresponding 2-D heat maps generated by (b) JSS, (c) SE [52], (d) BAM [54], and (e) CBAM [53] attention modules.

TABLE VI
ABLATION STUDY ON THE EFFECTS OF TL TOPOLOGY

	Baseline	TL-without LR	TL-with LR
OA(%)	85.45	90.81	91.68
AA(%)	86.13	91.05	91.64
κ	0.84	0.90	0.91

into 2-D heat maps by computing the mean of spatial positions along the channel axis that are displayed in Fig. 6(b)–(e).

As shown in Fig. 6, the attention mechanisms of SE, BAM, and CBAM tend to focus on similar spatial regions with varying scales. On the contrary, the JSS module results in an apparent different outcome. Comparing output heat maps in Fig. 6(b)–(e), it is demonstrated that the proposed pointwise attention mechanism, represented by heat map in Fig. 6(b), can more effectively discriminate similar spectral–spatial neighborhoods in the input than other counterparts.

3) Effects of TL Topology: Unlike the chain structure of the popular CNN frameworks, our proposed models incorporate residual and dense aggregations through a triangular 2-D lattice topology named TL. Here, we test the effectiveness of the proposed TL block. In particular, we adopt the plain network architecture by replacing the TL block with a standard residual CNN architecture consisting of six 2-D convolutional layers (named *conv1* to *conv6*) of 64, 64, 64, 96, 96, and 128 channels, respectively. Regarding residual links, the input of every convolutional layer is added to the output of the immediate next convolutional layer. We keep all other aspects (i.e., two-step bottleneck, attention, and classification blocks) of the baseline network the same as the proposed network for fair comparison. As shown in Table VI, the significant advantage of the proposed lattice block compared with the plain block is obvious. The TL block dramatically outperforms the baseline, even without residual aggregations, which suggests the effectiveness of the lattice topology to explore and exploit discriminative features.

Furthermore, we test the performance of the TL block in the presence and absence of LR aggregations. As expected, facilitating gradient flow with residual links within the lattice block can result in around 1% increase in the overall performance.

C. Classification Performance

To evaluate ALN's performance in HSI classification, we compare it with seven CNN-based benchmark methods. These methods include a variety of 2-D/3-D attention-based or nonattention-based CNN-topologies: a deep 2-D CNN model (ContextNet [69]), two 3-D CNN methods (Multiscale 3-D-CNN [21] and 3-D-CNN [22]), a ResNet-based framework (DPyResNet [42]), a spectral–spatial attention-based residual networks (SSRN [19]), an attention-based adaptive spectral–spatial kernel ResNet (A2S2K-ResNet [33]), and a spatial–spectral attention-aided CNN (SSAtt [24]). For ContextNet [69], Multiscale 3-D-CNN [21] and 3-D-CNN [22] implementations provided by DeepyperX toolbox[1] [70] are used and other models are reimplemented following their original articles. Performance metrics are evaluated on the Houston-2013, Indian Pines, and Pavia University datasets as explained in Section V-A. The overall classification performance in terms of the three quantitative objectives OA, AA, and κ alongside each class accuracy of the compared methods are reported in Tables VII–IX.

[1]https://gitlab.inria.fr/naudeber/DeepHyperX/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

COMPARISON OF THE CLASSIFICATION PERFORMANCE ON THE HOUSTON-2013 DATASET. THE CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE ARE REPORTED. THE BEST RESULTS ARE INDICATED IN BOLDFACE

Class No.	ContextNet [69]	Multi-scale 3D-CNN [21]	3D-CNN [22]	DPyResNet [42]	SSRN [19]	A2S2K-ResNet [33]	SSAtt [24]	ALN
1	90.58 ± 0.8	92.91 ± 1.0	**92.53** ± 3.5	89.50 ± 0.9	81.48 ± 1.6	90.50 ± 0.5	90.24 ± 0.6	88.60 ± 0.8
2	92.01 ± 1.8	**94.92** ± 1.0	94.71 ± 0.5	91.92 ± 2.5	92.48 ± 1.3	93.41 ± 3.1	91.91 ± 0.8	91.23 ± 0.6
3	86.35 ± 5.8	96.30 ± 0.9	97.10 ± 0.5	82.93 ± 3.0	98.02 ± 1.7	**99.51** ± 0.7	81.98 ± 6.9	87.16 ± 4.6
4	96.11 ± 1.8	97.13 ± 1.0	93.34 ± 7.4	94.80 ± 1.0	98.11 ± 1.0	95.58 ± 2.4	97.27 ± 0.6	**97.63** ± 0.8
5	93.42 ± 2.3	97.26 ± 1.6	95.60 ± 2.0	92.25± 4.4	**99.91** ± 1.6	92.29 ± 9.4	94.21 ± 2.1	96.88 ± 1.6
6	89.85 ± 5.4	77.29 ± 2.9	71.53 ± 5.4	79.86 ± 6.0	**95.80** ± 3.1	92.20 ± 4.1	79.81 ± 8.2	87.35 ± 6.1
7	82.78 ± 1.8	87.15 ± 1.5	82.94 ± 1.3	85.21 ± 2.1	89.46 ± 2.1	90.23 ± 2.4	**89.29**± 1.8	86.79 ± 2.1
8	74.64 ± 4.9	74.83 ± 3.5	73.76 ± 4.2	75.66 ± 6.5	69.90 ± 4.1	80.67 ± 8.6	76.64± 4.4	**84.56**± 5.5
9	76.65 ± 4.5	76.93 ± 3.4	74.06 ± 2.4	83.04 ± 3.1	84.04 ± 4.4	76.42 ± 8.8	84.19 ± 1.4	**90.48** ± 1.9
10	71.27 ± 7.1	60.62 ± 3.8	57.15 ± 4.8	72.67 ± 7.2	82.34 ± 5.9	58.13 ± 16.9	76.21 ± 4.4	**93.40** ± 4.5
11	84.22 ± 2.5	68.61 ± 2.1	65.73 ± 2.2	75.14 ± 4.1	93.17 ± 5.3	67.87 ± 11.7	81.43 ± 2.6	**97.12** ± 2.7
12	83.34 ± 4.3	74.52 ± 2.0	71.68 ± 5.8	79.17 ± 6.4	90.80 ± 3.7	75.28 ± 10.1	83.68 ± 1.4	**94.43** ± 1.4
13	**90.10** ± 1.9	88.32 ± 2.0	83.11 ± 10.5	80.38 ± 3.6	72.98 ± 5.6	87.58 ± 6.1	84.61 ± 2.6	87.85± 4.4
14	95.02 ± 2.5	93.97 ± 2.7	91.20 ± 3.2	86.97 ± 6.2	**99.19** ± 0.9	90.80 ± 9.4	90.44 ± 4.3	93.72 ± 3.0
15	97.19 ± 0.8	92.42 ± 2.2	91.64 ± 2.3	83.37 ± 4.6	94.08 ± 1.4	98.01 ± 1.5	92.57 ± 4.2	**97.54**± 2.6
OA(%)	85.68 ± 1.2	83.69 ± 1.1	81.57 ± 1.4	83.69 ± 2.2	89.46 ± 1.1	84.23 ± 4.2	86.36 ± 0.6	**91.68** ± 0.8
AA(%)	86.90 ± 1.3	84.76 ± 1.2	82.40 ± 1.6	83.52 ± 2.3	89.45 ± 1.1	85.87 ± 4.3	86.23 ± 0.7	**91.64** ± 0.9
κ	0.84 ± 0.014	0.82 ± 0.012	0.80 ± 0.016	0.82 ± 0.024	0.88 ± 0.012	0.83 ± 0.046	0.85 ± 0.007	**0.91** ± 0.009

TABLE VIII

COMPARISON OF THE CLASSIFICATION PERFORMANCE ON THE INDIAN PINES DATASET. THE CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE ARE REPORTED. THE BEST RESULTS ARE INDICATED IN BOLDFACE

Class No.	ContextNet [69]	Multi-scale 3D-CNN [21]	3D-CNN [22]	DPyResNet [42]	SSRN [19]	A2S2K-ResNet [33]	SSAtt [24]	ALN
1	19.77 ± 7.0	69.22 ± 21.5	45.90 ± 4.8	51.92 ± 22.2	80.91 ± 9.4	**82.32** ± 12.1	72.54 ± 12.6	81.82 ± 11.3
2	8.31 ± 10.8	62.20 ± 6.5	52.73 ± 5.0	31.81 ± 18.4	65.95 ± 5.3	82.50 ± 4.1	72.90 ± 4.6	**84.35** ± 5.6
3	15.24 ± 14.5	49.21 ± 2.9	40.54 ± 5.7	26.33 ± 13.7	54.05 ± 5.8	73.12 ± 4.2	59.35 ± 7.4	**75.86** ± 3.1
4	25.71 ± 10.4	64.82 ± 4.2	50.41 ± 9.1	33.42 ± 13.2	73.40 ± 9.2	79.00 ± 7.1	75.40 ± 7.0	**90.50** ± 4.4
5	17.51 ± 13.3	80.91 ± 5.0	71.23 ± 7.9	48.37 ± 17.7	81.93 ± 3.1	85.94 ± 2.9	83.84 ± 6.0	**87.88** ± 2.4
6	42.22 ± 27.4	90.30 ± 1.9	84.50 ± 5.1	70.01 ± 10.0	94.51 ± 1.6	96.93 ± 1.3	91.56 ± 3.5	**96.26** ± 2.2
7	29.52 ± 22.4	36.61 ± 14.1	23.58 ± 4.8	37.62 ± 21.5	50.39 ± 19.6	58.72 ± 24.2	75.32 ± 24.6	**78.18** ± 20.2
8	64.36 ± 22.6	94.46 ± 6.5	89.13 ± 3.3	82.15 ± 15.5	97.80 ± 1.5	99.34 ± 0.9	98.60 ± 1.1	**99.60** ± 0.4
9	17.81 ± 11.9	13.23 ± 3.5	14.55 ± 4.6	10.06 ± 6.7	**38.62** ± 21.0	26.67 ± 7.9	16.79 ± 10.1	30.79 ± 7.8
10	35.72 ± 19.1	65.10 ± 2.5	58.53 ± 3.5	38.74 ± 14.2	66.82 ± 4.6	73.61 ± 6.4	71.93 ± 5.9	**82.03** ± 4.1
11	7.10 ± 16.4	67.24 ± 4.4	59.60 ± 5.0	29.65 ± 25.7	72.11 ± 5.5	81.89 ± 3.8	71.91 ± 6.1	**83.75** ± 4.2
12	11.17 ± 7.5	53.15 ± 5.0	46.07 ± 5.4	34.41 ± 8.2	61.54 ± 4.8	77.34 ± 6.1	61.17 ± 7.0	**79.00** ± 6.5
13	60.72 ± 16.8	92.31 ± 5.5	87.93 ± 5.7	76.61 ± 12.8	93.90 ± 3.3	93.00 ± 2.5	93.24 ± 4.9	**90.53** ± 4.0
14	70.12 ± 25.2	93.52 ± 1.9	88.91 ± 3.1	78.52 ± 21.4	95.32 ± 2.7	97.30 ± 1.5	94.25 ± 1.5	**97.26** ± 1.2
15	26.01 ± 8.2	67.38 ± 4.7	53.28 ± 6.0	41.80 ± 8.9	68.41 ± 6.4	75.55 ± 4.0	62.20 ± 8.9	**76.17** ± 5.5
16	82.92 ± 21.0	95.63 ± 1.9	96.32 ± 1.9	69.63 ± 25.6	**92.36** ± 2.6	81.05 ± 6.5	84.33 ± 6.9	78.00 ± 7.8
OA(%)	32.10 ± 7.2	70.93 ± 2.3	62.56 ± 2.5	45.49 ± 11.2	73.73 ± 2.2	82.13 ± 1.7	75.01 ± 2.6	**84.34** ± 2.6
AA(%)	33.38 ± 7.6	68.45 ± 3.5	60.20 ± 2.7	47.56 ± 11.6	74.25 ± 2.3	78.97 ± 2.6	74.06 ± 2.7	**81.93** ± 2.5
κ	0.27 ± 0.065	0.65 ± 0.709	0.57 ± 0.027	0.40 ± 0.114	0.69 ± 0.024	0.799 ± 0.019	0.719 ± 0.030	**0.824** ± 0.029

TABLE IX

COMPARISON OF THE CLASSIFICATION PERFORMANCE ON THE PAVIA UNIVERSITY DATASET. THE CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE ARE REPORTED. THE BEST RESULTS ARE INDICATED IN BOLDFACE

Class No.	ContextNet [69]	Multi-scale 3D-CNN [21]	3D-CNN [22]	DPyResNet [42]	SSRN [19]	A2S2K-ResNet [33]	SSAtt [24]	ALN
1	67.72 ± 21.8	69.73 ± 8.0	82.70 ± 2.8	76.24 ± 4.4	89.41 ± 1.6	86.94 ± 12.6	88.32 ± 3.1	**92.82** ± 2.0
2	48.92 ± 20.4	79.90 ± 3.1	80.46 ± 6.2	74.81 ± 10.1	89.23 ± 3.2	85.13 ± 4.8	89.84 ± 2.5	**90.10** ± 7.0
3	29.01 ± 18.6	46.56 ± 10.4	70.15 ± 5.1	55.26 ± 6.5	75.94 ± 4.5	71.10 ± 13.6	74.25 ± 5.7	**87.86** ± 4.6
4	63.59 ± 9.4	87.03 ± 4.6	85.36 ± 7.0	88.79 ± 7.7	88.68 ± 8.5	86.12 ± 12.4	95.22 ± 1.1	**92.93** ± 1.9
5	99.10 ± 0.8	95.41 ± 3.8	99.60 ± 0.4	98.41 ± 1.3	**99.82** ± 0.2	99.80 ± 0.2	98.51 ± 1.0	99.40 ± 0.6
6	41.34 ± 4.6	53.33 ± 7.9	61.55 ± 5.1	56.37 ± 5.5	**88.56** ± 6.0	77.15 ± 9.6	81.93 ± 6.5	87.04 ± 9.9
7	44.57 ± 17.4	56.69 ± 8.4	69.71 ± 3.1	61.55 ± 5.8	82.75 ± 7.6	79.90 ± 17.3	80.80 ± 6.5	**89.85** ± 4.2
8	27.52 ± 26.4	60.75 ± 8.0	70.63 ± 3.4	70.11 ± 10.3	80.62 ± 2.5	81.00 ± 16.9	82.18 ± 6.8	**93.12** ± 3.2
9	87.20 ± 21.1	87.92 ± 8.9	95.42 ± 2.7	98.00 ± 1.0	99.00 ± 0.5	**99.40** ± 0.9	97.29 ± 1.8	98.00 ± 0.6
OA(%)	52.77 ± 10.9	67.84 ± 4.2	76.27 ± 3.9	70.70 ± 5.9	85.57 ± 2.8	81.93 ± 5.4	85.32 ± 2.6	**88.65** ± 4.5
AA(%)	56.55 ± 11.2	70.78 ± 5.3	79.50 ± 4.2	75.50 ± 6.5	88.22 ± 4.2	85.15 ± 6.7	87.56 ± 2.4	**92.37** ± 5.1
κ	0.44 ± 0.102	0.62 ± 0.051	0.69 ± 0.045	0.63 ± 0.061	0.81 ± 0.034	0.77 ± 0.064	0.81 ± 0.028	**0.85** ± 0.053

With respect to the Houston-2013 dataset, as shown in Table VII, the proposed ALN has a clear advantage over other benchmark methods. Specifically, the proposed ALN attains significant improvements of +5.36%, +7.45%, +2.22%, +7.99%, +10.11%, +7.99%, and +6.00% for OA over SSAtt [24], A2S2K-ResNet [33], SSRN [19], DPyResNet [42], 3-D-CNN [22], Multiscale 3-D-CNN [21], and ContextNet [69], respectively. Moreover, attention-based methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

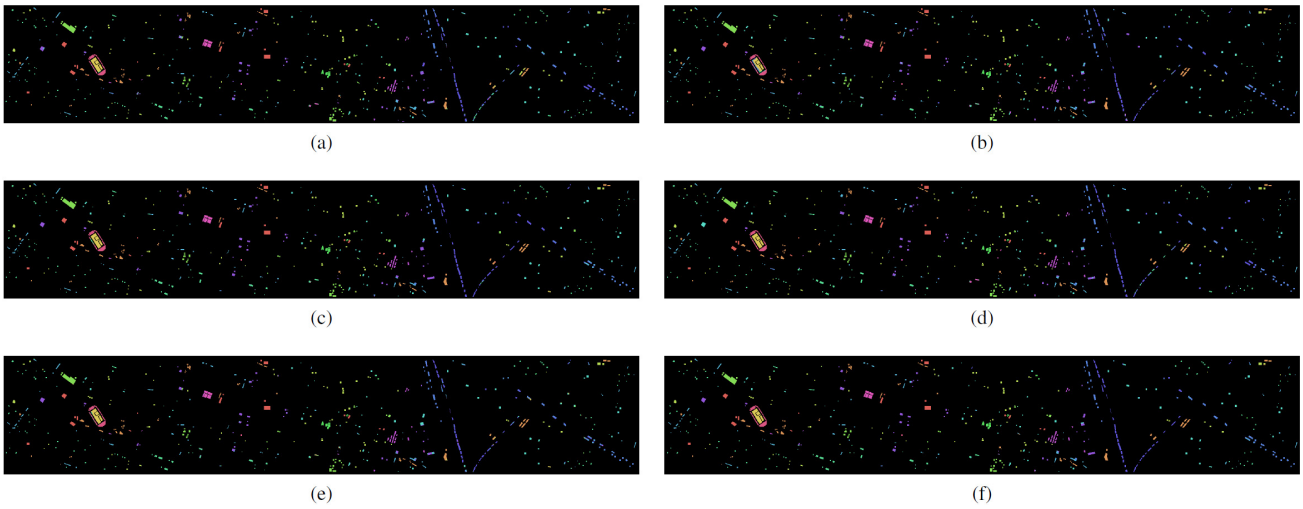


Fig. 7. Comparison of obtained classification maps on the "Houston-2013" dataset. (a) 3-D-CNN (84.79%). (b) DPyResNet (85.57%). (c) SSRN (89.59%). (d) A2S2K-ResNet (84.75%). (e) SSAtt (86.53%). (f) ALN (93.09%).

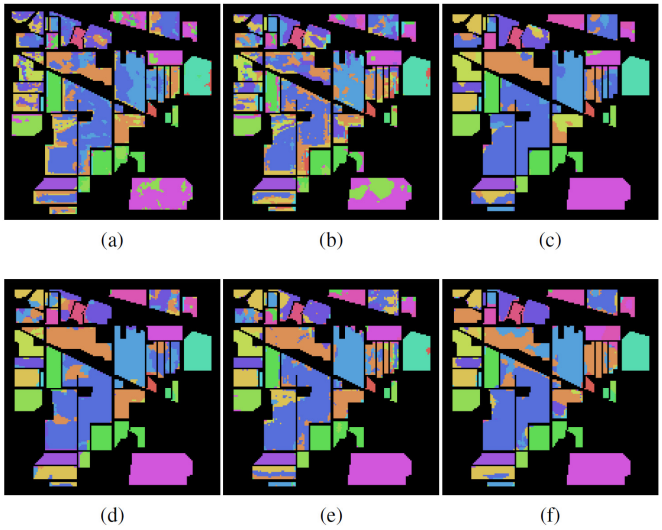


Fig. 8. Comparison of obtained classification maps on the Indian Pines dataset. (a) 3-D-CNN (60.31%). (b) DPyResNet (56.60%). (c) SSRN (82.11%). (d) A2S2K-ResNet (82.33%). (e) SSAtt (78.17%). (f) ALN (84.56%).

such as SSAtt [24], A2S2K-ResNet [33], and SSRN [19] are unable to achieve relatively high accuracy ($\geq$80%) for each category. For instance, the SSAtt [24] and A2S2K-ResNet [33] obtain the low accuracy of 58.13% and 76.21% for the Highway category (class No. 10), respectively. Likewise, SSRN [19] model achieves the poor performance of 69.90% for the Commercial category (class No. 8). This phenomenon is worse for methods such as DPyResNet [42], 3-D-CNN [22], Multiscale 3-D-CNN [21], and ContextNet [69], so that their spectral discrimination power seems to be biased for some specific categories. On the other hand, the proposed ALN offers a good generalization capability to produce superior and/or competitive classification accuracy for each class compared with other methods.

Besides quantitative comparison, Fig. 7 illustrates the visual comparison of classification maps generated by a single

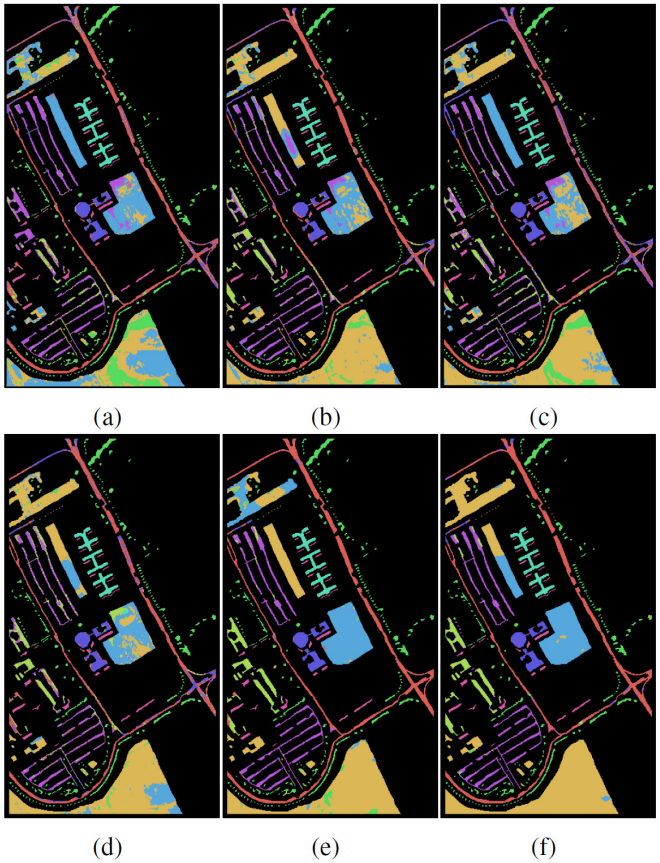


Fig. 9. Comparison of obtained classification maps on the Indian Pines dataset. (a) 3-D-CNN (68.53%). (b) DPyResNet (74.75%). (c) SSRN (87.63%). (d) A2S2K-ResNet (85.71%). (e) SSAtt (85.20%). (f) ALN (89.50%).

experiment of different methods on the Houston-2013 dataset. The proposed ALN demonstrates more consistency with the ground-truth map (see Fig. 3) than others.

In addition to Houston-2013 dataset, Table VIII lists the classification performance obtained by different models on the Indian Pines dataset. The lower spatial resolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

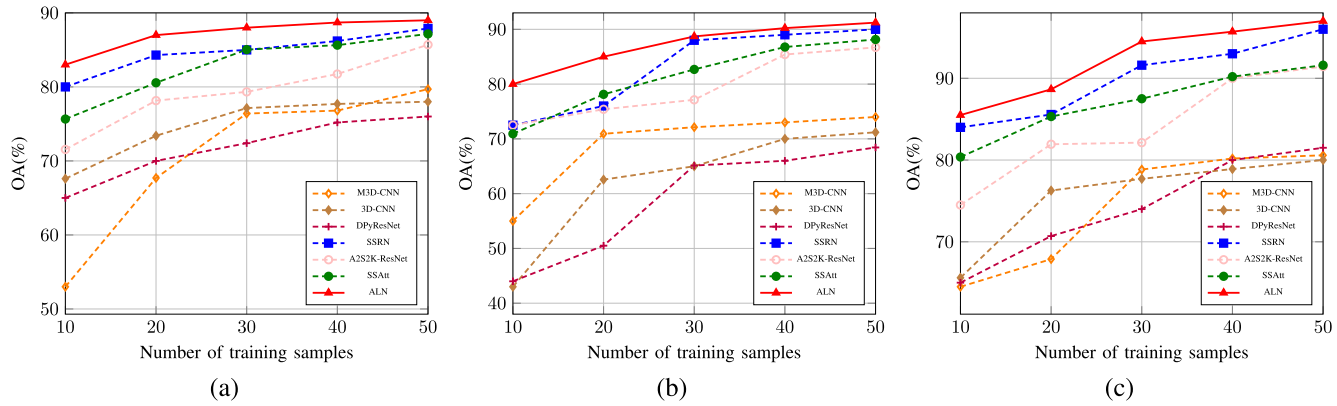


Fig. 10. OA(%) achieved by different methods with varying the number of training samples. (a) Houston-2013. (b) Indian Pines. (c) Pavia University.

TABLE X

COMPARISON OF MODEL SIZE (IN MILLION PARAMETERS), FLOPS (IN MILLION FLOPS), OA, AND COMPUTATION TIME (IN SECONDS) OF DIFFERENT ADVANCED METHODS FOR HSI CLASSIFICATION. OA VALUES ARE TESTED ON HOUSTON-2013 DATASET

Methods	ContextNet [69]	Multi-scale 3D-CNN [21]	3D-CNN [22]	DPyResNet [42]	SSRN [19]	A2S2K-ResNet [33]	SSAtt [24]	ALN
# params	0.31M	0.41M	0.04M	22.21M	0.28M	0.37M	0.32M	0.20M
# FLOPs	25.80M	31.56M	49.87M	38.18M	112.67M	340.11M	7.52M	9.81M
Training (s)	223.14	314.99	51.64	272.29	1750.73	1163.48	289.18	136.73
Test (s)	25.80	65.19	10.24	91.75	125.48	105.04	44.98	42.90
OA (%)	85.68	83.69	81.57	83.69	89.46	82.13	86.36	91.68

(20 m per pixel) of this dataset causes a high spectral mixing issue compared with other studied datasets. Moreover, the numbers of class samples in this dataset are quite unbalanced. In particular, the numbers of class samples in Alfalfa, Grass-pasture-mowed, and Oats are very small. Therefore, the compared methods' overall performance on this dataset is relatively lower than others. However, the proposed ALN outperforms other counterparts significantly in OA, AA, and κ coefficient. The apparent gap between ALN and the other benchmark methods that use similar techniques such as residual aggregation (DPyResNet [42]) and spectral/spatial attention mechanism (SSRN [19], A2S2K-ResNet [33], and SSAtt [24]) demonstrate the effectiveness of the proposed JSS attention-based framework. Moreover, the qualitative comparison of all the methods shown in Fig. 8 verifies the clear advantage of our new attention-based lattice topology over other ones.

Furthermore, we use Pavia University to evaluate the generalization of the proposed ALN model on the HSI classification task. Table IX and Fig. 9, respectively, show the quantitative and qualitative classification performance of eight methods when only 20 labeled samples per class are used for training. Unlike Indian Pines, the overall performance of all models is greatly improved because of the higher spatial resolution of the Pavia University dataset (1.3 m per pixel). Moreover, it can be seen that our ALN model exhibits the best results in most cases, demonstrating the generalization ability of the proposed ALN on a different dataset. More specifically, compared with the second-best model (i.e., SSRN [19]), ALN can provide the noticeable increase in the classification accuracy of 11.92% and 12.5% on Gravel land cover (Class No. 3) and Bricks category (Class No. 8), respectively.

D. Effect of Training Samples

We further investigate the effect of the number of training samples on the proposed method and other counterparts. In particular, we evaluate the performance of different methods with varying sample sizes from 10 to 50 for Houston-2013, Indian Pines, and Pavia University datasets, respectively. The curves of OA with respect to different numbers of training set sizes are shown in Fig. 10. As expected, when a larger training set is used, the performances of all methods increase. However, the proposed ALN outperforms other methods on all three datasets. Using only ten training samples per class, ALN has achieved the OA of more than 83%, 80%, and 85% on Houston-2013, Indian Pines, and Pavia University datasets, respectively. This indicates the robustness of the proposed ALN when a sufficiently small number of training samples are used. It can also be seen from Fig. 10 that the attention-based methods demonstrate significantly higher performance than the nonattention-based counterparts, which verifies the effect of the spectral/spatial feature recalibration mechanism to extract more discriminative features.

E. Model Complexity Analysis

We also conduct a comparison of the model complexity in terms of model size, computational cost, overall performance, and running time between the proposed ALN and other advanced CNN-based approaches for HSIs' classification task as listed in Table X. The runtime, including training time and test time for all compared methods, is measured by adopting a Titan RTX GPU and reimplementing them following their original articles. Although involving more computation is the inevitable cost of utilizing multiple convolutional elements,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

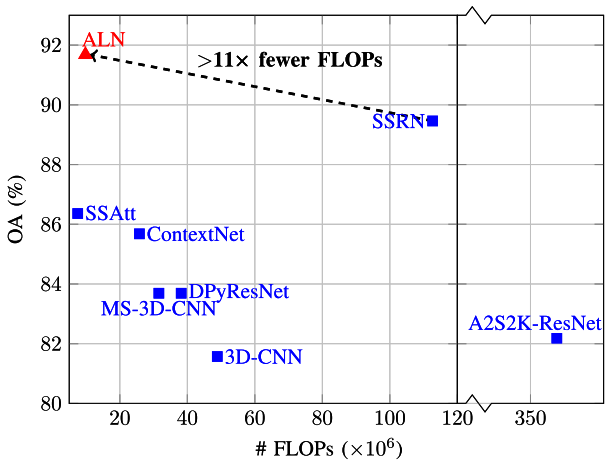


Fig. 11. Comparison of model computation cost (i.e., the number of FLOPs) against classification performance [OA (%)] on the Houston-2013 dataset.

ALN presents a great superiority and efficiency in model complexity objectives. On the other hand, it can be seen that 3-D CNN-based methods such as Multiscale 3-D-CNN [21], 3-D-CNN [22], and SSRN [19] have substantially high computation costs. Meanwhile, 3-D-CNN [22], the lightest model (0.04 M parameters), shows the lowest training time and test time because of the simplicity of its structure. Nevertheless, it is considerably outperformed by other methods in terms of classification performance. Compared with SSAtt [24] and A2S2K-ResNet [33] which utilize both spatial and spectral attention mechanisms, ALN can obtain a higher OA value of 2.22% and 5.32% with better efficiency concerning model size and training/testing time, respectively. We further visualize the computational cost [the number of floating point operations (FLOPs)] and performance comparison in Fig. 11. Though techniques such as attention mechanism and residual aggregations have also been adopted in SSRN [19], the ALN that requires 9.81 million FLOPs achieved an OA value much higher than that of the SSRN [19] that requires 11.48× as many FLOPs. This shows that ALN employs these techniques effectively and efficiently so that an excellent performance for both model complexity criteria and accuracy of the HSI classification is obtained.

VI. Conclusion

In this article, we present a novel ALN for HSI classification. Unlike other CNNs that use either residual or dense aggregations, ALN takes advantage of both aggregation types more efficiently and effectively. Moreover, ALN utilizes a unique pointwise spectral schema alongside spatial attention mechanisms to effectively exploit interdependences among spectral and spatial features. These properties enable ALN to achieve a superior performance and an excellent efficiency in terms of model complexity than 2-D/3-D CNN-based benchmarks in HSI classification. We further verify the effectiveness of each component of the proposed ALN via an extensive ablation study. Further improvements on the ALN topology and investigating its capabilities in other remote sensing applications are interesting future works.

Acknowledgment

The authors would like to thank N. Audebert *et al.* [70] for providing DeepHyperX toolbox that can be used to benchmark different deep learning methods on various public hyperspectral datasets.

References

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.
[2] S. Delalieux, P. J. Zarco-Tejada, L. Tits, M. Á. J. Bello, D. S. Intriglolo, and B. Somers, "Unmixing-based fusion of hyperspatial and hyperspectral airborne imagery for early detection of vegetation stress," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2571–2582, Jun. 2014.
[3] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, 2004.
[4] D. L. Donoho *et al.*, "High-dimensional data analysis: The curses and blessings of dimensionality," *AMS Math Challenges Lect.*, vol. 1, p. 32, Aug. 2000.
[5] M. Huang, Q. Zhu, B. Wang, and R. Lu, "Analysis of hyperspectral scattering images using locally linear embedding algorithm for apple mealiness classification," *Comput. Electron. Agriculture*, vol. 89, pp. 175–181, Nov. 2012.
[6] X. Liu, X. Feng, F. Liu, and Y. He, "Identification of hybrid Rice strain based on near-infrared hyperspectral imaging technology," *Trans. Chin. Soc. Agricult. Eng.*, vol. 33, no. 22, pp. 189–194, 2017.
[7] J. Li, Y. Zhang, M. Liu, J. Chen, and L. Xue, "Rapid detection and visualization of mechanical bruises on 'Nanfeng' Mandarin using the hyperspectral imaging combined with ICA_LSQ method," *Food Anal. Methods*, vol. 12, no. 9, pp. 2025–2034, 2019.
[8] S. Jia, Y. Xie, G. Tang, and J. Zhu, "Spatial-spectral-combined sparse representation-based classification for hyperspectral imagery," *Soft Comput.*, vol. 20, no. 12, pp. 4659–4668, Dec. 2016.
[9] L. Fang, C. Wang, S. Li, and J. A. Benediktsson, "Hyperspectral image classification via multiple-feature-based adaptive sparse representation," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 7, pp. 1646–1657, Jul. 2017.
[10] S. Jia, B. Deng, J. Zhu, X. Jia, and Q. Li, "Local binary pattern-based hyperspectral image classification with superpixel guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 749–759, Feb. 2018.
[11] S. Jia, X. Deng, J. Zhu, M. Xu, J. Zhou, and X. Jia, "Collaborative representation-based multiscale superpixel fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7770–7784, Oct. 2019.
[12] S. Jia, J. Hu, J. Zhu, X. Jia, and Q. Li, "Three-dimensional local binary patterns for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2399–2413, Apr. 2017.
[13] S. Jia, Z. Lin, B. Deng, J. Zhu, and Q. Li, "Cascade superpixel regularized Gabor feature fusion for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1638–1652, May 2019.
[14] J. Gu *et al.*, "Recent advances in convolutional neural networks," *Pattern Recognit.*, vol. 77, pp. 354–377, May 2018.
[15] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. ECCV*. Springer, 2016, pp. 630–645.
[16] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. CVPR*, Jul. 2017, pp. 4700–4708.
[17] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.
[18] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral-spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.
[19] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

[20] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[21] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[22] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[23] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, 2017.

[24] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[25] M. Nikzad, A. Nicolson, Y. Gao, J. Zhou, K. K. Paliwal, and F. Shang, "Deep residual-dense lattice network for speech enhancement," in *Proc. AAAI*, 2020, pp. 8552–8559.

[26] L. Zhu, R. Deng, M. Maire, Z. Deng, G. Mori, and P. Tan, "Sparsely aggregated convolutional networks," in *Proc. ECCV*, 2018, pp. 186–201.

[27] M. Nikzad, Y. Gao, and J. Zhou, "Attention-based pyramid dilated lattice network for blind image denoising," in *Proc. 13th Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 931–937.

[28] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[29] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[30] C. Zhao, X. Wan, G. Zhao, B. Cui, W. Liu, and B. Qi, "Spectral-spatial classification of hyperspectral imagery based on stacked sparse autoencoder and random forest," *Eur. J. Remote Sens.*, vol. 50, no. 1, pp. 47–63, 2017.

[31] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[32] P. Liu, H. Zhang, and K. B. Eom, "Active deep learning for classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 2, pp. 712–724, Feb. 2017.

[33] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[34] F. I. Alam, J. Zhou, A. W.-C. Liew, X. Jia, J. Chanussot, and Y. Gao, "Conditional random field and deep feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1612–1628, Mar. 2019.

[35] S. Jia *et al.*, "A lightweight convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4150–4163, May 2021.

[36] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[37] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[38] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. NeurIPS*, 2012, pp. 1097–1105.

[39] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015, pp. 1–15.

[40] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. CVPR*, Jun. 2015, pp. 1–9.

[41] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. ECCV*. Springer, 2014, pp. 818–833.

[42] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Aug. 2019.

[43] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, 2018.

[44] X. Kang, B. Zhuo, and P. Duan, "Dual-path network-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 447–451, Mar. 2019.

[45] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, "Deep networks with stochastic depth," in *Proc. ECCV*. Springer, 2016, pp. 646–661.

[46] W. Czaja and M. Ehler, "Schroedinger eigenmaps for the analysis of biomedical data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 5, pp. 1274–1280, May 2013.

[47] M. Corbetta and G. L. Shulman, "Control of goal-directed and stimulus-driven attention in the brain," *Nature Rev. Neurosci.*, vol. 3, no. 3, pp. 201–215, Mar. 2002.

[48] X. Zhang, J. C. Park, J. Salant, S. Thomas, J. Hirsch, and D. C. Hood, "A multiplicative model for spatial interaction in the human visual cortex," *J. Vis.*, vol. 8, no. 8, p. 4, Jun. 2008.

[49] K. Xu *et al.*, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. ICML*, 2015, pp. 2048–2057.

[50] S. Yohanandan, A. Song, A. G. Dyer, and D. Tao, "Saliency preservation in low-resolution grayscale images," in *Proc. ECCV*, 2018, pp. 235–251.

[51] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3156–3164.

[52] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.

[53] S. Woo, J. Park, J.-Y. Lee, and I. So Kweon, "CBAM: Convolutional block attention module," in *Proc. ECCV*, 2018, pp. 3–19.

[54] J. Park, S. Woo, J.-Y. Lee, and I. S. Kweon, "BAM: Bottleneck attention module," in *Proc. BMVC*, 2018, pp. 1–14.

[55] J. Wang, J. Zhou, and W. Huang, "Attend in bands: Hyperspectral band weighting and selection for image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4712–4727, Dec. 2019.

[56] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, May 2020.

[57] S. K. Roy, S. R. Dubey, S. Chatterjee, and B. B. Chaudhuri, "FuSENet: Fused squeeze-and-excitation network for spectral-spatial hyperspectral image classification," *IET Image Process.*, vol. 14, no. 8, pp. 1653–1661, 2020.

[58] B. Fang, Y. Li, H. Zhang, and J. C.-W. Chan, "Hyperspectral images classification based on dense convolutional networks with spectral-wise attention mechanism," *Remote Sens.*, vol. 11, no. 2, p. 159, 2019.

[59] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[60] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. ICML*, 2010, pp. 807–814.

[61] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[62] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[63] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. AISTATS*, 2011, pp. 315–323.

[64] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[65] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2014, pp. 1–11.

[66] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. ICLR*, 2016, pp. 1–13.

[67] C. Debes *et al.*, "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[68] J. Liang, J. Zhou, Y. Qian, L. Wen, X. Bai, and Y. Gao, "On the sampling strategy for evaluation of spectral-spatial methods in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 862–880, Feb. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.

[69] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.
[70] N. Audebert, B. L. Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.



Mohammad Nikzad (Student Member, IEEE) received the B.Sc. degree in computer engineering from Yazd University, Yazd, Iran, in 2005, the M.Sc. degree in artificial intelligence from Azad Science and Research University, Tehran, Iran, in 2010, and the Ph.D. degree in artificial intelligence from Griffith University, Nathan, QLD, Australia, in 2021.

He is a Research Fellow with the Institute for Integrated and Intelligent Systems, Griffith University. His research interests include deep learning, machine learning, digital image processing, and computer vision.



Yongsheng Gao (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electronic engineering from Zhejiang University, Hangzhou, China, in 1985 and 1988, respectively, and the Ph.D. degree in computer engineering from Nanyang Technological University, Singapore, in 2000.

He is a Professor with the School of Engineering and Built Environment, Griffith University, Nathan, QLD, Australia, and the Director of ARC Research Hub for Driving Farming Productivity and Disease Prevention, Australia. He had been the Leader of Biosecurity Group, Queensland Research Laboratory, National ICT Australia (ARC Center of Excellence), a Consultant of Panasonic Singapore Laboratories, and an Assistant Professor with the School of Computer Engineering, Nanyang Technological University. His research interests include smart farming, intelligent agriculture, biosecurity, face recognition, biometrics, image retrieval, computer vision, pattern recognition, environmental informatics, and medical imaging.



Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from the Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

He was a Research Fellow with the Research School of Computer Science, The Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. He is an Associate Professor with the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia. His research interests include pattern recognition, computer vision, and spectral imaging with their applications to remote sensing and environmental informatics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:03:36 UTC from IEEE Xplore. Restrictions apply.